Remy - Data and Database

Urban Mobility Data Explorer | NYC yellow taxi, Jan 2019 | Timezone America/New_York, local strings kept as-is.

Owner: Remy. Files: database/, scripts/, backend/features.js. Manuce owns backend/db.js and the API. Kenny owns frontend/.

1. Files delivered

- database/schema.sql: taxi_zones, data_imports, trips, view trip_features
- database/inspection.md: R01 source record
- database/final-scope.md: R07 scope and counts
- scripts/setup-db.js: create database and run schema
- scripts/inspect-sources.js: stream headers and row counts
- scripts/import-zones.js: load 265 zones, join geometry by LocationID
- scripts/clean-trip.js: cleanTrip reject/flag rules and fingerprint
- backend/features.js: toSeconds, getDurationMinutes, getSpeedMph, getFarePerMile
- scripts/check-features.js: 10 known-answer checks, all pass
- scripts/import-trips.js: streaming batched import with audit
- scripts/verify-db.js: R06 join, feature, filter and EXPLAIN checks

2. R01 - Source inspection

yellow_tripdata_2019-01.csv

NYC TLC yellow trip records. 18 columns: VendorID, pickup, dropoff, passenger_count, trip_distance, RatecodeID, store_fwd_flag, PULocationID, DOLocationID, payment_type, fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, total_amount, congestion_surcharge.

Sample: VendorID 1, pickup 2019-01-01 00:46:40, dropoff 2019-01-01 00:53:20, 1 passenger, 1.50 miles, PU 151, DO 239, fare 7, total 9.95.

20,000-row sample: distance min 0, max 128.73, zero 150, over 100 only 1. Fare min -52, max 496.5, negative 18. Bad dropoff-before-pickup 5 in 5000.

taxi_zone_lookup.csv

Columns: LocationID, Borough, Zone, service_zone. Rows: 265, IDs 1-265, no duplicate IDs. Duplicate names: Corona twice (56, 57), Governor's Island three times (103, 104, 105). IDs 264 (Unknown) and 265 (Outside NYC) have no polygon.

taxi_zones shapefile

Folder taxi_zones (.shp .dbf .shx .prj). ID property is LocationID. 263 shapes, 260 unique IDs. IDs 56 and 103 appear twice (split polygons, merged to MultiPolygon). Missing polygons: 57, 104, 105, 264, 265. Converted to taxi_zones.geojson with 260 features. Join rule: integer LocationID. Lookup rows without polygons are kept with NULL geometry.

3. R02 - Database schema

- taxi_zones: location_id PK, borough, zone_name, service_zone, geometry LONGTEXT NULL.
- data_imports: source, checksum, scope, cleaning v1, status running/complete/failed, rows_read, normal, flagged, excluded, period_start/end.
- trips: source fields, import_id FK, pickup/dropoff zone FKs, fingerprint CHAR(64), is_flagged, flag_reason.
- trip_features VIEW: duration_min, speed_mph, fare_per_mile.
- UNIQUE (import_id, fingerprint) blocks exact duplicates.

- Indexes: pickup_datetime (date filters), pickup_zone_id and dropoff_zone_id (map and route grouping).

4. R03 - Zone import

scripts/import-zones.js reads the CSV with a quoted-field parser and the GeoJSON file, joins by integer LocationID, and upserts with REPLACE. It prints zone count, duplicate IDs, and missing-polygon count. Expected: zones 265, missing_polygon 5. Valid lookup rows without polygons are preserved.

5. R04 - Cleaning rules and features

- reject missing_time: pickup or dropoff empty or unparsable.
- reject bad_time: dropoff minus pickup <= 0 seconds.
- reject missing_value: distance or fare empty.
- reject negative_value: distance < 0 or fare < 0.
- reject unknown_zone: PU or DO id not in taxi_zones.
- flag zero_distance: distance == 0, kept for analysis.
- flag outlier: distance > 100 or fare > 500 or duration > 1440 min or speed > 100 mph.

Features. Duration = positive seconds / 60. Speed = miles x 3600 / positive seconds, NULL when distance <= 0. Fare per mile = fare / miles, NULL when distance <= 0.

Known trip: 2019-01-01 00:46:40 to 00:53:20, 1.5 miles, fare 7 gives duration 6.67 min, speed 13.5 mph, fare per mile 4.67. Zero distance gives fare per mile NULL. check-features.js covers accept, bad time, negative, zero, outlier, unknown zone, and the three formulas.

6. R05 - Trip import

- Streaming read with csv-parse, bounded by --limit (default 5000), batches of --batch (default 500).
- Fingerprint = sha256(pickup|dropoff|PU|DO|distance|fare|total). Duplicate insert is counted as excluded.
- Rerun rule: same checksum + scope with status complete returns the old import id and inserts nothing.
- Failure rule: any error marks the import failed; dashboard reads only status complete.
- Demo result on 5000 rows: read 5000, normal 4952, flagged 34, excluded 14 (bad_time 5, negative_value 9).

7. R06 - Verification with Manuce

- Trips count for the import equals normal + flagged in data_imports.
- No trips with missing zone joins (LEFT JOIN NULL check = 0).
- trip_features row matches backend/features.js for the known trip.
- Date filter uses pickup >= day AND pickup < next day; EXPLAIN shows idx_trips_pickup_time in use.
- Run: npm run verify-db.

8. R07 - Final scope and limits

Demo scope is the first 5000 rows (early 2019-01-01 trips), not the full month. Full month needs a high-limit run and more time. Thresholds are v1 from the sample above. Unexpected: negative fares exist (e.g. -52) and some zero-distance trips carry a normal fare, so they are flagged and kept rather than deleted.

9. Handoff to Manuce

- Schema and names: database/schema.sql.
- Setup: npm install; npm run setup-db; npm run import-zones; npm run import-trips -- --limit=5000 --batch=500.
- Active import: latest row with status complete; period from period_start/period_end; quality from rows_read/normal/flagged/excluded.
- Known trip for query check: PU 151 to DO 239, duration 6.67, speed 13.5, fare/mi 4.67.